Office hour

TA Office Hours will be available from Week 3 to Week 16.

Feel free to drop by and ask any questions!

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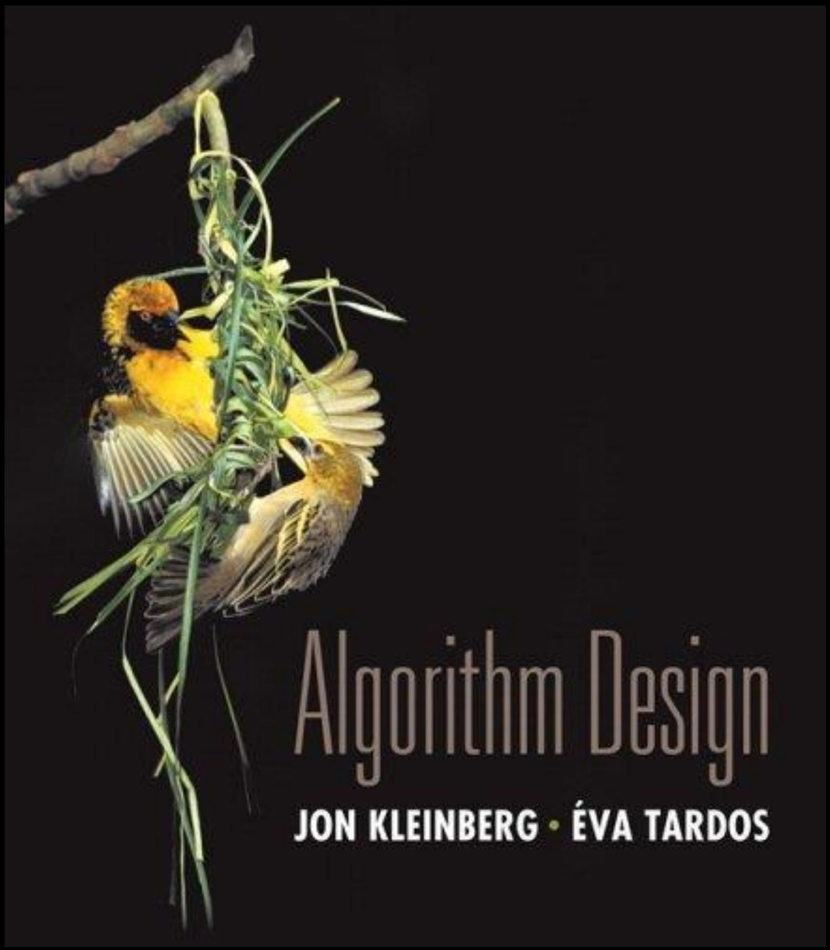
Thursday	10:00-11:00	顾书齐	信息学院 1A-307	gushq2024@
Thursday	17:00-18:00	何好雨	信息学院 2-407	hehy12024@



Chapter 5

Divide and Conquer

Divide-and-Conquer

Divide-and-conquer.

- Break up problem into several parts.
- Solve each part recursively.
- Combine solutions to sub-problems into overall solution.

Most common usage.

- Break up problem of size n into two equal parts of size $\frac{1}{2}n$ in linear time.
- Solve two parts recursively.
- Combine two solutions into overall solution in linear time.

Consequence.

- Divide-and-conquer: $\Theta(n \log n)$

Selection

Given list A of n numbers, and $i \leq n$, Select(A, i) finds i'th smallest number in A.

- Ex $A = [3,1,6,7,2]$, Select(A,4)=6.
- Assume numbers all distinct, for simplicity.

Select generalizes median.

- Median of A is Select($A, n/2$).

We can solve select by first sorting the numbers in $O(n \log n)$ time, then choosing the i'th largest one.

We show how to solve Select, and hence median, in $O(n)$ time.

Select algorithm overview

Break A into many small groups.
Find the median of each small group directly.
Recursively find the median of this set of medians (using Select).
Use the median-of-medians to partition A.
Keep looking for target in one of the two partitions.

Select algorithm 1

Say A has n items, and we want i'th largest.

- Assume for simplicity n divides 5.

Divide A into $n/5$ groups of 5 elements each.

- First 5 elements of A in first group, next 5 in second group, etc.

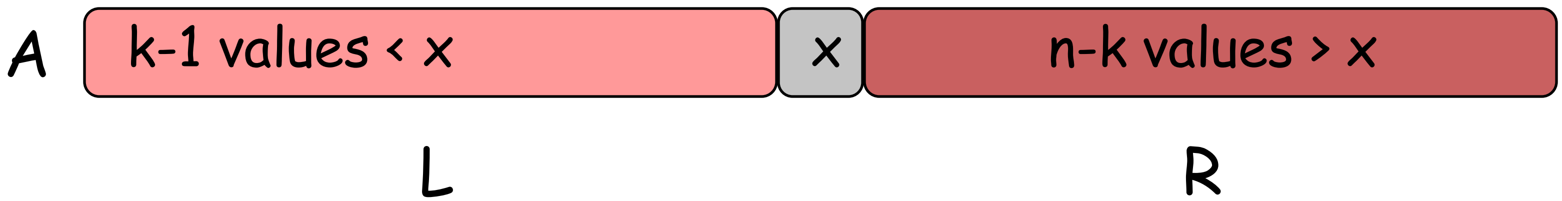
For each group of 5 elements, find its median.

- E.g. sort the 5 numbers, take 3rd value.
- Let B be the set of $n/5$ medians.

Recursively find the median in B.

- I.e. do Select($B, n/10$), finding median in B.
- Say the median is x.

Select algorithm 2



Partition A using x as pivot.

- Move all values $< x$ to the left of x, all other values to the right.
- If A has n values, this takes $O(n)$ time.

Say there are k values $\leq x$ in A, and $n - k$ values $> x$.

If $i = k$, return x.

- We want the i'th largest item in A, which is x.

If $i < k$, return Select(L, i).

- Ex If $i = 5$ and $k = 10$, then fifth largest item in A is fifth largest item in L.

Else $i > k$, return Select$(R, i - k)$.

- Ex if $i = 15$ and $k = 10$, then fifteenth largest item in A is fifth largest item in R.

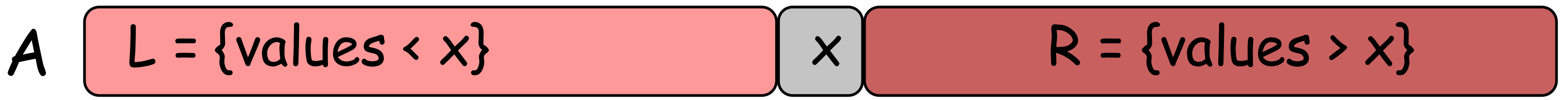
Select analysis 1

Let $S(n)$ be an upper bound on select's running time given a list of size n.

$$S(n) = S(n/5) + S(u) + O(n)$$

- u is the size of whichever partition we recurse on.
- $O(n)$ term comes from two parts.
  - First, for each of the $n/5$ groups of 5 numbers, find their median in constant time $\Rightarrow O(n)$ time overall.
  - Next, partition array in $O(n)$ time.
- $S(n/5)$ to find the median of the $n/5$ group medians.
- The key to ensuring select runs fast is ensuring u is small.
- We will show $u \leq 7n/10$.

Select analysis 2



Let x be the median of medians.

We show that $|L| \geq 3n/10$, and $|R| \geq 3n/10$.

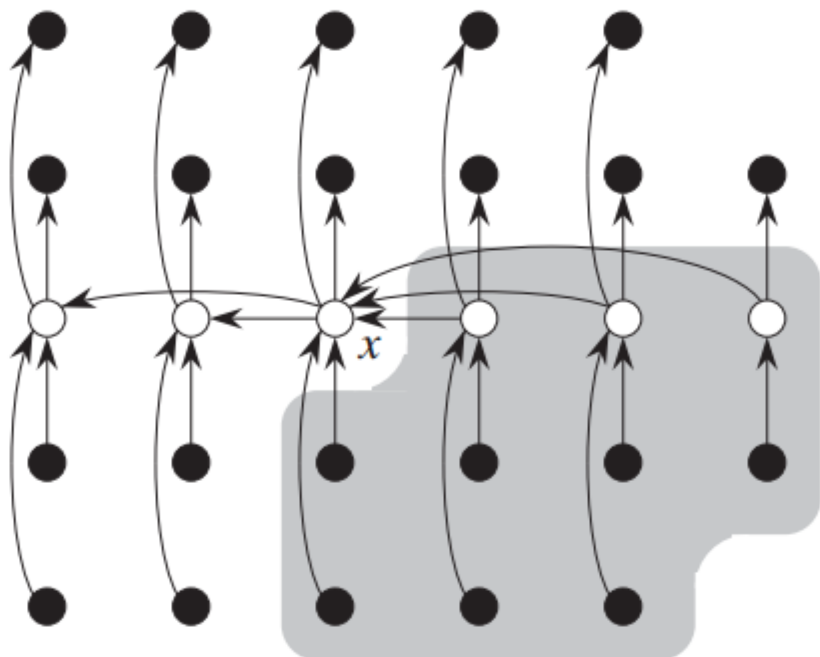
- I.e. there are at least 3n/10 values less than x, and 3n/10 values larger than x.

Then, also get $|L| \leq 7n/10$.

- $|L| = n - 1 - |R|$.
- Also, $|R| \leq 7n/10$.

Thus, $S(n) \leq S(n/5) + S(7n/10) + O(n)$.

Select analysis 3



- Groups of 5 elements are shown in columns.
- Medians are shown in white.
- Medians less than x are shown to the right.

Source: *Introduction to Algorithms*
Cormen, Leiserson, Rivest, Stein

There are $n/5 * 1/2 = n/10$ medians less than x.

For each such median, there are two more values from the group less than the median.

- There are 3 values from each group < x.

So there are at least $3n/10$ values < x.

I.e. $|L| \geq 3n/10$.

Similarly, $|R| \geq 3n/10$.

Select analysis 4

We have $S(n) \leq S(n/5) + S(7n/10) + O(n)$.

- Since we want to upper bound $S(n)$, we let $S(n) = S(n/5) + S(7n/10) + O(n)$.
- For sufficiently large n, the $O(n)$ term is bounded by bn, for some constant b.

We guess $S(n) = O(n)$, and show this satisfies equation above, i.e. the guess is valid.

Since $S(n) = O(n), S(n) \leq cn$ for sufficiently large n, for some constant c.

Select analysis 5

From $S(n) = S(n/5) + S(7n/10) + bn$ and $S(n) = cn$, we get

$cn = c(n/5) + c * 7n/10 + bn.$

So $cn = 9cn/10 + bn$, and $c = 10b$.

Thus, $S(n) \le 10bn = O(n)$ for sufficiently large n, for some constant b.

5.5 Integer Multiplication

Integer Arithmetic

Add. Given two n-digit integers a and b, compute a + b.

- O(n) bit operations.

Multiply. Given two n-digit integers a and b, compute a × b.

- Brute force solution: $\Theta(n^2)$ bit operations.

	1	1	0	1	0	1	0	1
+	0	1	1	1	1	1	0	1
1	0	1	0	1	0	0	1	0

Add

Multiply

								1	1	0	1	0	1	0	1	0
								*	0	1	1	1	1	1	0	1
								1	1	0	1	0	1	0	1	0
							0	0	0	0	0	0	0	0	0	
						1	1	0	1	0	1	0	1	0		
					1	1	0	1	0	1	0	1	0			
				1	1	0	1	0	1	0	1	0				
			1	1	0	1	0	1	0	1	0					
		1	1	0	1	0	1	0	1	0						
	0	0	0	0	0	0	0	0	0							
0	1	1	0	1	0	0	0	0	0	0	0	0	0	0	1	0

Divide-and-Conquer Multiplication: Warmup

To multiply two n-digit integers:
- Multiply four $\frac{1}{2}$n-digit integers.
- Add two $\frac{1}{2}$n-digit integers, and shift to obtain result.

$$x = \underbrace{1\,0\,0\,0}_{x_1}\ \underbrace{1\,1\,0\,1}_{x_0}$$

$$\begin{aligned}
x &= 2^{n/2} \cdot x_1 + x_0 \\
y &= 2^{n/2} \cdot y_1 + y_0 \\
xy &= \left(2^{n/2} \cdot x_1 + x_0\right)\left(2^{n/2} \cdot y_1 + y_0\right) = 2^n \cdot x_1 y_1 + 2^{n/2} \cdot \left(x_1 y_0 + x_0 y_1\right) + x_0 y_0
\end{aligned}$$

$$\mathrm{T}(n) = \underbrace{4T(n/2)}_{\text{recursive calls}} + \underbrace{\Theta(n)}_{\text{add, shift}} \Rightarrow \mathrm{T}(n) = \Theta(n^2)$$

assumes n is a power of 2

Multiplying complex numbers

Compute $(a + bi)(c + di) = x + yi$.

- i is the imaginary number, i.e. $i^2 = -1$.

$$x = ac - bd, y = ad + bc$$

- 4 multiplications, 2 additions.

Can we do better?

Yes! From Gauss, we have $x = ac - bd, y = (a + b)(c + d) - ac - bd$.

- 3 multiplications: $ac, bd, (a + b)(c + d)$.
- 5 additions.

Why does Gauss's method matter?

- It's useful when addition faster than multiplication.
- It can be used recursively to speed up integer multiplication.

Karatsuba multiplication

$\mathbf{a} = \underbrace{1000}_{a_1}\underbrace{1101}_{a_0}$ $\qquad$ $\mathbf{b} = \underbrace{1110}_{b_1}\underbrace{0001}_{b_0}$

$$(a + bi)(c + di) = x + yi$$
$$x = ac - bd$$
$$y = (a + b)(c + d) - ac - bd$$

$$\begin{aligned}
a &= 2^{n/2} \cdot a_1 + a_0 \\
b &= 2^{n/2} \cdot b_1 + b_0 \\
ab &= 2^{n} \cdot a_1 b_1 + 2^{n/2} \cdot \left(a_1 b_0 + a_0 b_1\right) + a_0 b_0 \\
&= 2^{n} \cdot a_1 b_1 + 2^{n/2} \cdot \left((a_1 + a_0)(b_1 + b_0) - a_1 b_1 - a_0 b_0\right) + a_0 b_0
\end{aligned}$$

Karatsuba multiplication rearranges the terms in divide and conquer multiplication.

It does 3 multiplications of $n/2$ digit numbers instead of 4.

Applied recursively, this makes all the difference!

Notice the similarity between Karatsuba's and Gauss's method.

Karatsuba complexity

$$\begin{aligned} a &= 2^{n/2} \cdot a_1 + a_0 \\ b &= 2^{n/2} \cdot b_1 + b_0 \\ ab &= 2^n \cdot a_1 b_1 + 2^{n/2} \cdot (a_1 b_0 + a_0 b_1) + a_0 b_0 \\ &= 2^n \cdot a_1 b_1 + 2^{n/2} \cdot ((a_1 + a_0)(b_1 + b_0) - a_1 b_1 - a_0 b_0) + a_0 b_0 \end{aligned}$$

S(n) = time to multiply two n digit numbers using Karatsuba.

3 multiplications of $n/2$ digit numbers.

- $a_0 b_0, a_1 b_1, (a_1 + a_0)(b_1 + b_0)$.
- Takes $3S(n/2)$ time.

Multiply by 2^n and $2^{n/2}$.

- Done by shifting in $O(n)$ time.
- Ex $2^4 * 1011_2 = 10110000_2$.

6 additions / subtractions of $2n$ digit numbers.

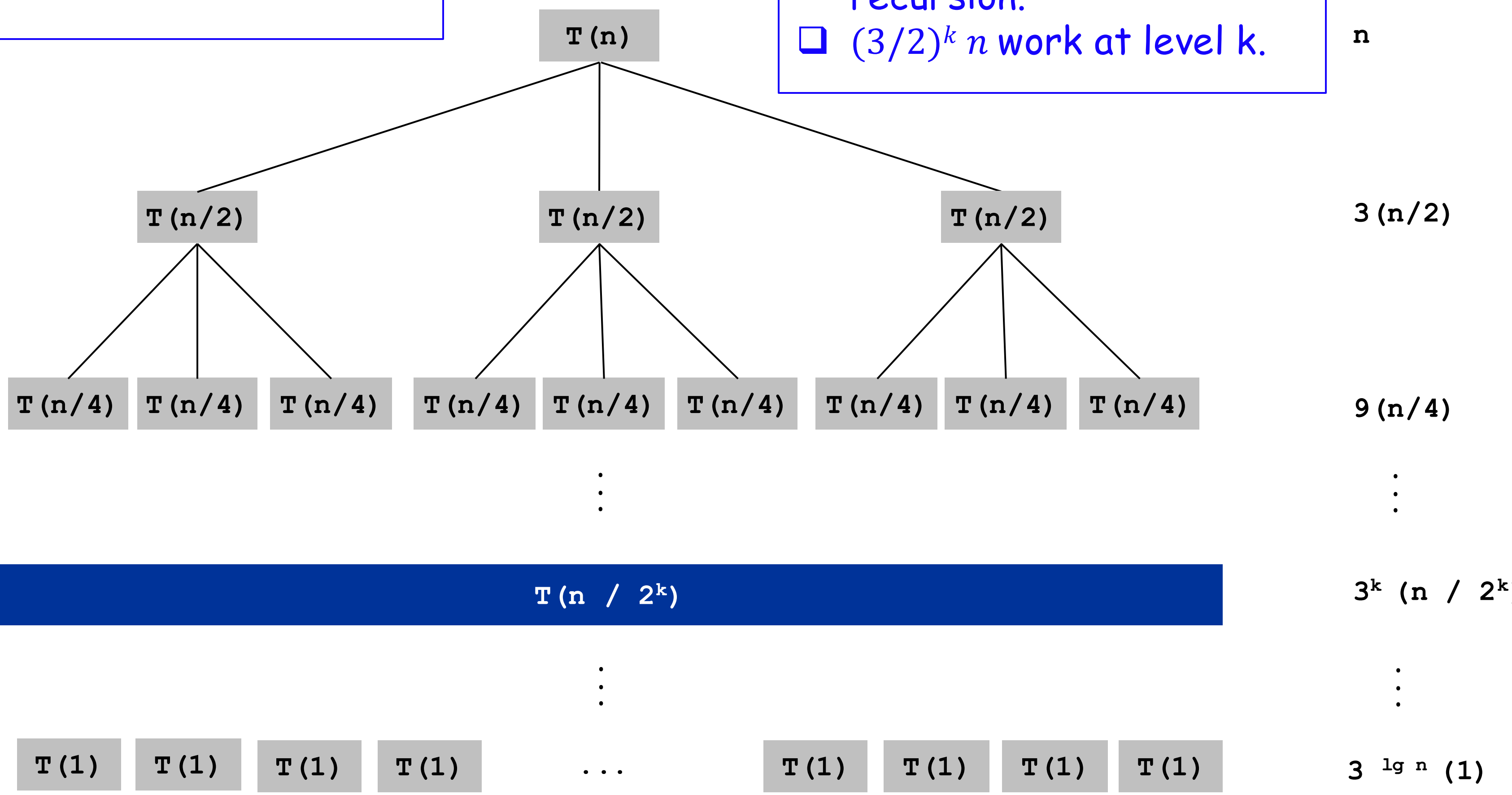
- $O(n)$ time.

$$S(n) = 3S(n/2) + O(n).$$
$$S(1) = 1.$$

Karatsuba recursion tree

$S(n) = 3S(n/2) + O(n)$

- $\lg n = \log_2 n$ levels of recursion.
- $(3/2)^k n$ work at level k.



Source: http://www.cs.bu.edu/fac/byers/courses/330/S13/handouts/05multiply.ppt

Karatsuba complexity

- $S(n) = n + \left(\frac{3}{2}\right)n + \left(\frac{3}{2}\right)^2 n + \cdots + \left(\frac{3}{2}\right)^{\lg n} n$
- $S(n) = n\left(\frac{\left(\frac{3}{2}\right)^{1+\lg n} - 1}{\frac{3}{2} - 1}\right) = 2n\left(\frac{3n^{\lg 3}}{2 \cdot 2^{\lg n}} - 1\right) = 3n^{\lg 3} - 2n$
  - $\lg 3 \approx 1.59.$
- So Karatsuba's method runs in $O(n^{1.59})$ instead of $O(n^2)$ time!
  - Practical for moderate n > 100.
- Useful for e.g. RSA cryptography.
- Even faster methods exist.
  - Schonhage-Strassen's Fast Fourier Transform based algorithm runs in $O(n \log n \log \log n)$ time.
  - Only practical for large n (> 10,000).

Karatsuba Multiplication

To multiply two n-digit integers:

- Add two $\frac{1}{2}$n digit integers.
- Multiply three $\frac{1}{2}$n-digit integers.
- Add, subtract, and shift $\frac{1}{2}$n-digit integers to obtain result.

$$
\begin{aligned}
x &= 2^{n/2} \cdot x_1 + x_0 \\
y &= 2^{n/2} \cdot y_1 + y_0 \\
xy &= 2^{n} \cdot x_1 y_1 + 2^{n/2} \cdot \left(x_1 y_0 + x_0 y_1\right) + x_0 y_0 \\
&= 2^{n} \cdot \underbrace{x_1 y_1}_{A} + 2^{n/2} \cdot \left(\underbrace{(x_1 + x_0)(y_1 + y_0)}_{B} - \underbrace{x_1 y_1}_{A} - \underbrace{x_0 y_0}_{C}\right) + \underbrace{x_0 y_0}_{C}
\end{aligned}
$$

Theorem. [Karatsuba-Ofman, 1962] Can multiply two n-digit integers in $O(n^{1.585})$ bit operations.

$$
\mathrm{T}(n) \le \underbrace{T\left(\lfloor n/2 \rfloor\right) + T\left(\lceil n/2 \rceil\right) + T\left(1 + \lceil n/2 \rceil\right)}_{\text{recursive calls}} + \underbrace{\Theta(n)}_{\text{add, subtract,shift}}
$$

$$
\Rightarrow \mathrm{T}(n) = O(n^{\log_2 3}) = O(n^{1.585})
$$

Matrix Multiplication

Complexity of matrix multiplication

$$\begin{bmatrix} c_{11} & c_{12} & \cdots & c_{1n} \\ c_{21} & c_{22} & \cdots & c_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ c_{n1} & c_{n2} & \cdots & c_{nn} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \times \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ b_{21} & b_{22} & \cdots & b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{nn} \end{bmatrix}$$

$$\begin{bmatrix} .59 & .32 & .41 \\ .31 & .36 & .25 \\ .45 & .31 & .42 \end{bmatrix} = \begin{bmatrix} .70 & .20 & .10 \\ .30 & .60 & .10 \\ .50 & .10 & .40 \end{bmatrix} \times \begin{bmatrix} .80 & .30 & .50 \\ .10 & .40 & .10 \\ .10 & .30 & .40 \end{bmatrix}$$

Source: http://www.cs.bu.edu/fac/byers/courses/330/S13/handouts/05multiply.ppt

```
for (i=0; i<n; i++) {
  for (j=0; j<n; j++) {
    C[i,j] = 0;
    for (k=0; k<n; k++)
      C[i,j] = C[i,j] + A[i,k]*B[k,j]
```

- Multiplying two $n \times n$ matrices the naive way takes $\Theta(n^3)$ time.
- Can we do better?
- Yes! Use Strassen's algorithm (1969).

Block matrix multiplication

C_{11} A_{11} A_{12} B_{11} B_{21}

$$\begin{bmatrix} 152 & 158 & 164 & 170 \\ 504 & 526 & 548 & 570 \\ 856 & 894 & 932 & 970 \\ 1208 & 1262 & 1316 & 1370 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 2 & 3 \\ 4 & 5 & 6 & 7 \\ 8 & 9 & 10 & 11 \\ 12 & 13 & 14 & 15 \end{bmatrix} \times \begin{bmatrix} 16 & 17 & 18 & 19 \\ 20 & 21 & 22 & 23 \\ 24 & 25 & 26 & 27 \\ 28 & 29 & 30 & 31 \end{bmatrix}$$

$$C_{11} = A_{11} \times B_{11} + A_{12} \times B_{21} = \begin{bmatrix} 0 & 1 \\ 4 & 5 \end{bmatrix} \times \begin{bmatrix} 16 & 17 \\ 20 & 21 \end{bmatrix} + \begin{bmatrix} 2 & 3 \\ 6 & 7 \end{bmatrix} \times \begin{bmatrix} 24 & 25 \\ 28 & 29 \end{bmatrix} = \begin{bmatrix} 152 & 158 \\ 504 & 526 \end{bmatrix}$$

Break A, B and C into blocks. Multiply them as in normal matrix multiplication.

Each block can be broken into subblocks and multiplied same way.

Leads to recursive matrix multiplication method.

Block matrix multiplication

$$\begin{bmatrix} 152 & 158 & 164 & 170 \\ 504 & 526 & 548 & 570 \\ 856 & 894 & 932 & 970 \\ 1208 & 1262 & 1316 & 1370 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 2 & 3 \\ 4 & 5 & 6 & 7 \\ 8 & 9 & 10 & 11 \\ 12 & 13 & 14 & 15 \end{bmatrix} \times \begin{bmatrix} 16 & 17 & 18 & 19 \\ 20 & 21 & 22 & 23 \\ 24 & 25 & 26 & 27 \\ 28 & 29 & 30 & 31 \end{bmatrix}$$

$$C_{11} = A_{11} \times B_{11} + A_{12} \times B_{21} = \begin{bmatrix} 0 & 1 \\ 4 & 5 \end{bmatrix} \times \begin{bmatrix} 16 & 17 \\ 20 & 21 \end{bmatrix} + \begin{bmatrix} 2 & 3 \\ 6 & 7 \end{bmatrix} \times \begin{bmatrix} 24 & 25 \\ 28 & 29 \end{bmatrix} = \begin{bmatrix} 152 & 158 \\ 504 & 526 \end{bmatrix}$$

S(n) = time for $n \times n$ block matrix multiplication.

C_{11} requires two $(n/2) \times (n/2)$ block matrix multiplications $\Rightarrow 2S(n/2)$ time.

C_{12}, C_{21} and C_{22} also require $2S(n/2)$ time each.

We also add four pairs of $(n/2) \times (n/2)$ matrices.

- Takes $O(n^2)$ time.

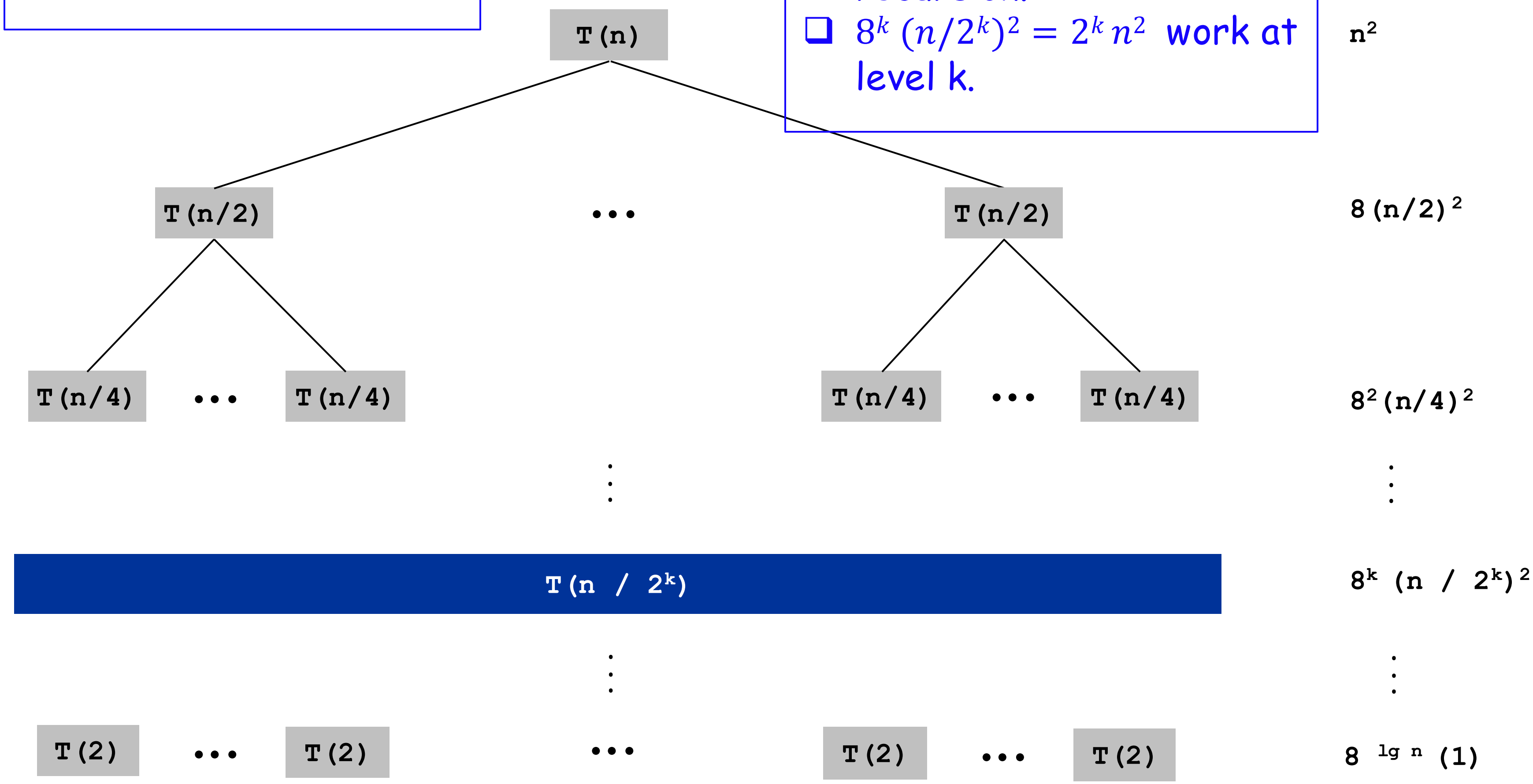
So $S(n) = 8S(n/2) + O(n^2)$.

- $S(2) = O(1)$.

Block MM recursion tree

$S(n) = 8S(n/2) + O(n^2)$

- $\lg n = \log_2 n$ levels of recursion.
- $8^k\,(n/2^k)^2 = 2^k\,n^2$ work at level k.



Block MM complexity

- $S(n) = n^2 + 2n^2 + 4n^2 + \ \ldots + 2^{\lg n} n^2 \ = n^2\,(1 + 2 + \cdots + 2^{\lg n}) = n^2(2n - 1) = \Theta(n^3)$.
- So simple block matrix multiplication takes same time as naive matrix multiplication.
- Problem is each recursion does 8 $(n/2) \times (n/2)$ MMs.
- Strassen's algorithm does 7 $(n/2) \times (n/2)$ MMs in each recursion.
  - Complexity becomes $O(n^{\lg_2 7}) = O\ (n^{2.81})$.

Matrix Multiplication: Key Idea

Key idea. multiply 2-by-2 block matrices with only 7 multiplications.

$$\begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \times \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}$$

$$\begin{aligned} C_{11} &= P_5 + P_4 - P_2 + P_6 \\ C_{12} &= P_1 + P_2 \\ C_{21} &= P_3 + P_4 \\ C_{22} &= P_5 + P_1 - P_3 - P_7 \end{aligned}$$

$$\begin{aligned} P_1 &= A_{11} \times (B_{12} - B_{22}) \\ P_2 &= (A_{11} + A_{12}) \times B_{22} \\ P_3 &= (A_{21} + A_{22}) \times B_{11} \\ P_4 &= A_{22} \times (B_{21} - B_{11}) \\ P_5 &= (A_{11} + A_{22}) \times (B_{11} + B_{22}) \\ P_6 &= (A_{12} - A_{22}) \times (B_{21} + B_{22}) \\ P_7 &= (A_{11} - A_{21}) \times (B_{11} + B_{12}) \end{aligned}$$

- 7 multiplications.
- 18 = 10 + 8 additions (or subtractions).

Strassen's algorithm

$$\begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \times \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}$$

$$\begin{aligned} C_{11} &= P_5 + P_4 - P_2 + P_6 \\ C_{12} &= P_1 + P_2 \\ C_{21} &= P_3 + P_4 \\ C_{22} &= P_5 + P_1 - P_3 - P_7 \end{aligned}$$

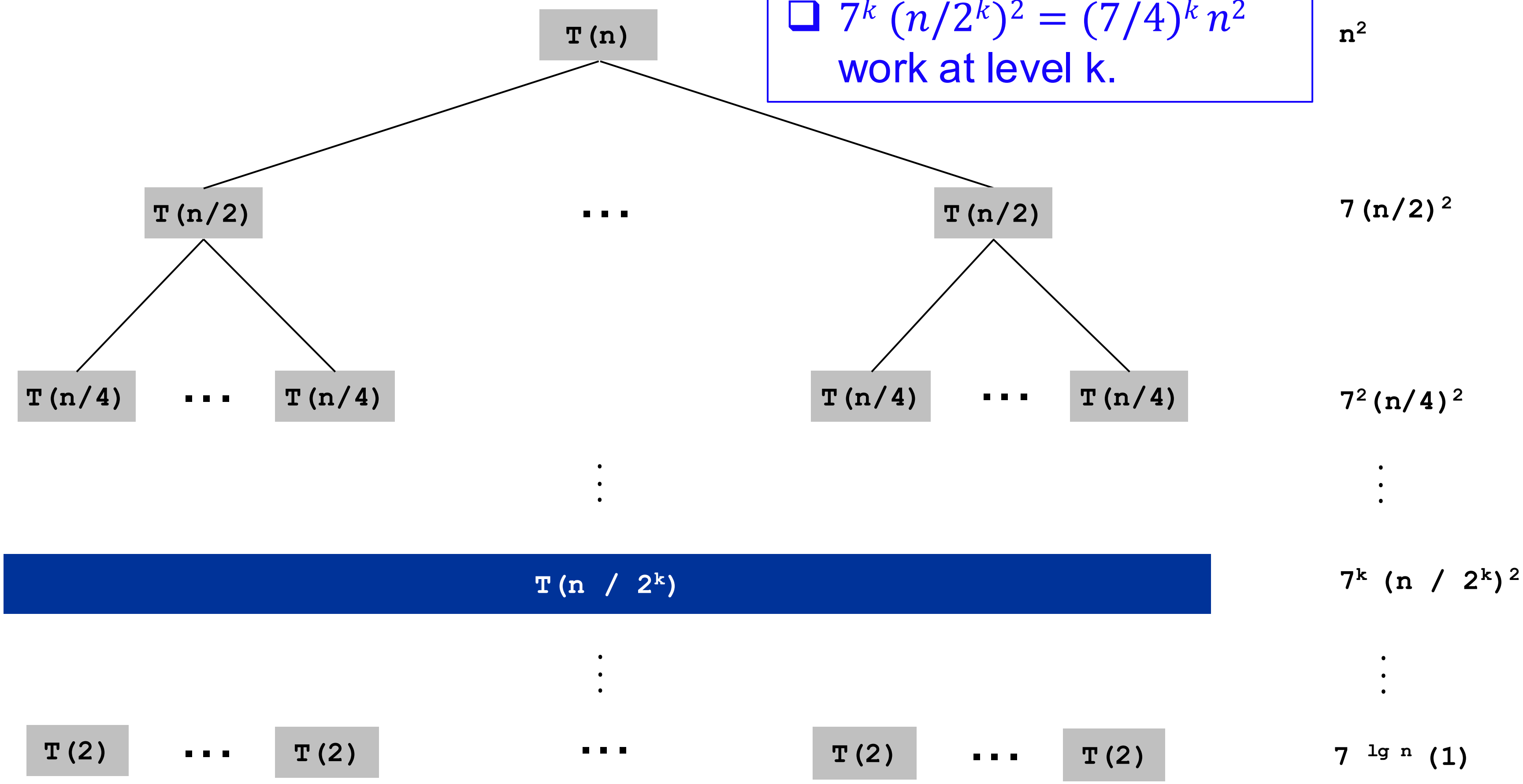
$$\begin{aligned} P_1 &= A_{11} \times (B_{12} - B_{22}) \\ P_2 &= (A_{11} + A_{12}) \times B_{22} \\ P_3 &= (A_{21} + A_{22}) \times B_{11} \\ P_4 &= A_{22} \times (B_{21} - B_{11}) \\ P_5 &= (A_{11} + A_{22}) \times (B_{11} + B_{22}) \\ P_6 &= (A_{12} - A_{22}) \times (B_{21} + B_{22}) \\ P_7 &= (A_{11} - A_{21}) \times (B_{11} + B_{12}) \end{aligned}$$

- Strassen's algorithm does 7 multiplications and 18 additions / subtractions of $(n/2) \times (n/2)$ matrices.
- Let S(n) be Strassen's algorithm's complexity.
- $S(n) = 7S(n/2) + O(n^2)$.
  - $O(n^2)$ comes from the matrix addition/subtractions.
- S(2) = O(1).

Strassen recursion tree

$S(n) = 7S(n/2) + O(n^2)$

- $\lg n = \log_2 n$ levels of recursion.
- $7^k\,(n/2^k)^2 = (7/4)^k\,n^2$ work at level k.



More about matrix multiplication

- Strassen's algorithm was a huge surprise. Before Strassen, it was widely believed matrix multiplication required $\Omega(n^3)$ time.
- Not clear how Strassen discovered the algorithm. Maybe inspiration from Karatsuba?
- Strassen's algorithm is practical. Beats naive method for $n > 20 - 1000$, depending on hardware architecture.
- Since Strassen, more sophisticated (but impractical) algorithms with $O(n^{2.375})$ time discovered.
- Some conjecture matrix multiplication can be done in $O(n^{2+\epsilon})$ time, for any $\epsilon > 0$.
  - This is nearly optimal, since even writing out $n \times n$ matrix output requires $O(n^2)$ time.

Fast Matrix Multiplication in Practice

Common misperception: "Strassen is only a theoretical curiosity."

- Advanced Computation Group at Apple Computer reports 8x speedup on G4 Velocity Engine when n ~ 2,500.
- Range of instances where it's useful is a subject of controversy.

Remark. Can "Strassenize" Ax=b, determinant, eigenvalues, and other matrix ops.

Fast Matrix Multiplication in Theory

Q. Multiply two 2-by-2 matrices with only 7 scalar multiplications?

A. Yes! [Strassen, 1969]

$$\Theta(n^{\log_2 7}) = O(n^{2.81})$$

Q. Multiply two 2-by-2 matrices with only 6 scalar multiplications?

A. Impossible. [Hopcroft and Kerr, 1971]

$$\Theta(n^{\log_2 6}) = O(n^{2.59})$$

Q. Two 3-by-3 matrices with only 21 scalar multiplications?

A. Also impossible.

$$\Theta(n^{\log_3 21}) = O(n^{2.77})$$

Q. Two 70-by-70 matrices with only 143,640 scalar multiplications?

A. Yes! [Pan, 1980]

$$\Theta(n^{\log_{70} 143640}) = O(n^{2.80})$$

Fast Matrix Multiplication in Theory

Decimal wars.

- December, 1979: $O(n^{2.521813})$.
- January, 1980: $O(n^{2.521801})$.
- …
- 1987: $O(n^{2.375477})$.
- 2010: $O(n^{2.374})$.
- 2011: $O(n^{2.3728642})$.
- 2014: $O(n^{2.3728639})$.

Best known. $O(n^{2.3728639})$ [François Le Gall, 2014.]

Conjecture. $O(n^{2+\varepsilon})$ for any $\varepsilon > 0$.

Caveat. Theoretical improvements to Strassen are progressively less practical.